

5. Sam carries out an experiment to determine the mass of a metal block using the principle of moments.

(a) State the principle of moments.

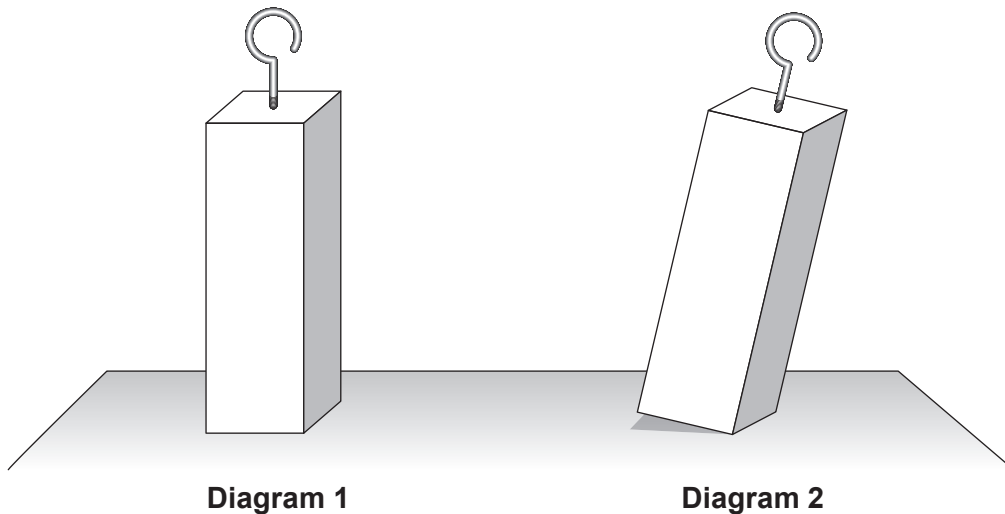
[2]

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- (b) The metal block is in the form of a cuboid as shown in **Diagram 1**. When setting up the experiment Sam accidentally knocks the metal block so that it tilts to the position shown in **Diagram 2**. The block then returns to the upright position as in **Diagram 1**.



Explain, in terms of centre of gravity, why the block returns to the upright position.

[2]

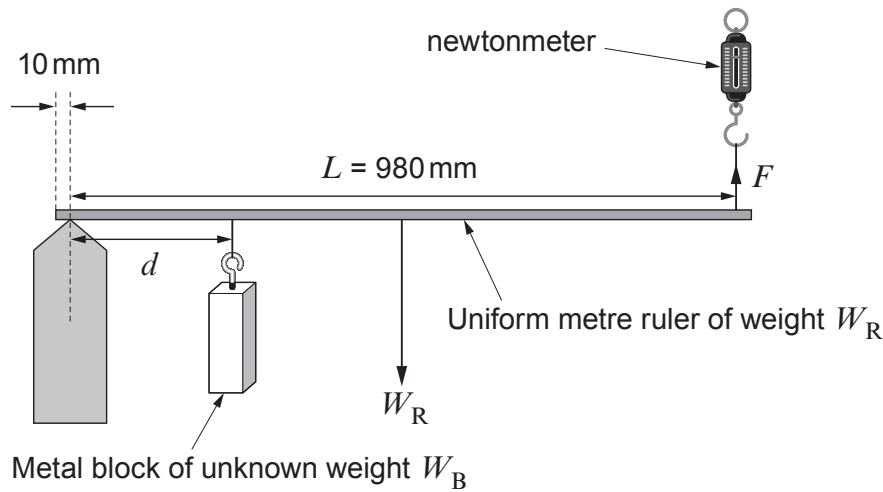
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(c) Sam sets up the experiment as shown using a metre ruler.



Sam uses the newtonmeter to measure the force,  $F$ , needed to make the metre ruler horizontal for varying values of  $d$ . He carries out the experiment twice and determines a mean value for  $F$  for each of the distances  $d$ . Sam judges that the ruler is horizontal by eye before taking readings for  $F$ . The measurements are recorded in the table below.

| Distance, $d$ / mm | Force, $F$ / N |         |      |
|--------------------|----------------|---------|------|
|                    | Trial 1        | Trial 2 | Mean |
| 100                | 2.8            | 2.6     |      |
| 250                | 5.2            | 5.7     |      |
| 400                | 8.8            | 8.5     |      |
| 550                | 11.2           | 11.6    |      |
| 700                | 14.6           | 14.5    |      |
| 850                | 17.2           | 17.6    |      |

- (i) Complete the table. [1]
- (ii) The following relationship is used to find the weight,  $W_B$ , of the block:

$$F = \frac{W_B d}{L} + \frac{W_R}{2}$$

By taking moments about the pivot, show how this relationship is obtained. [2]

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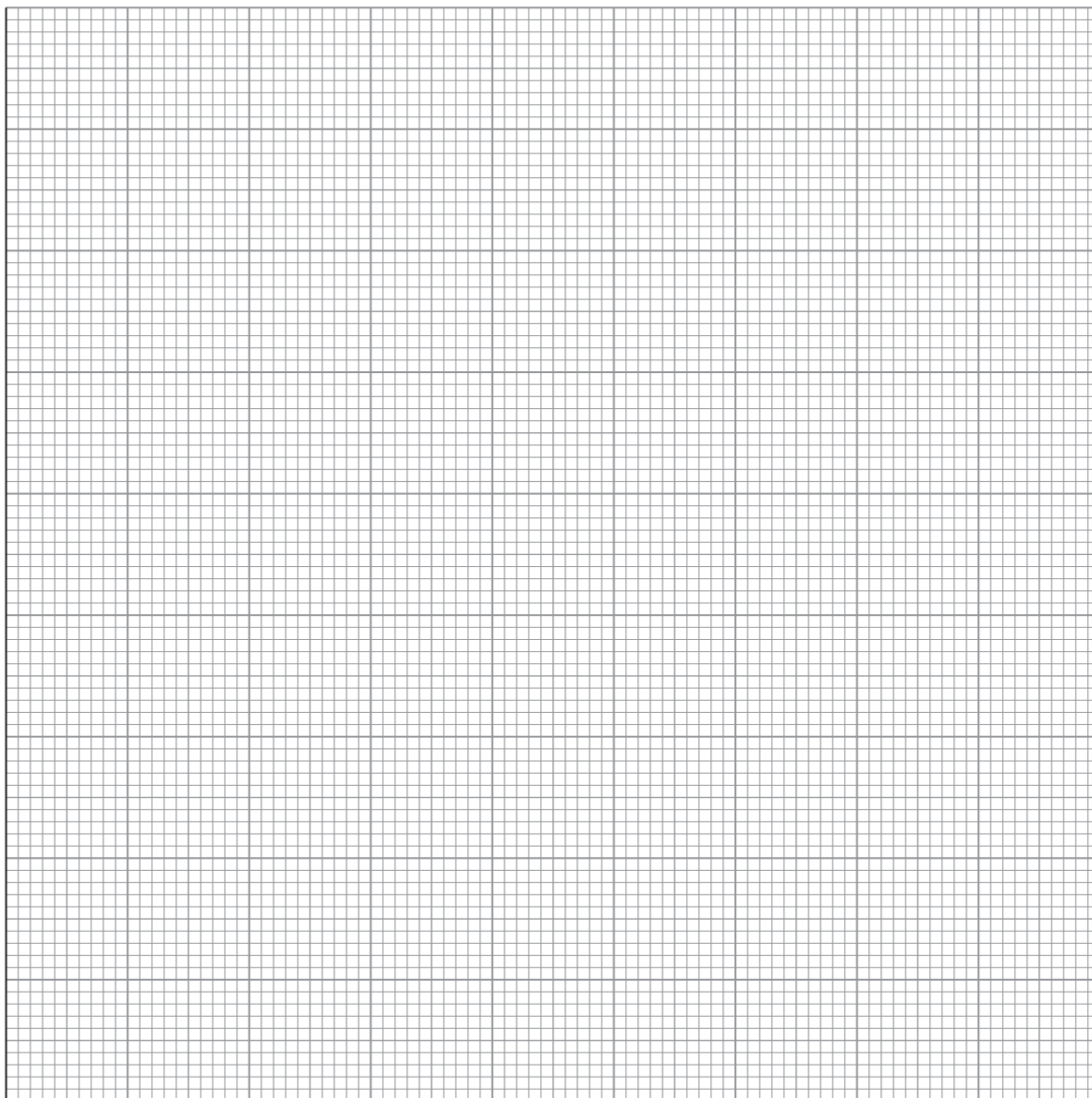
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- (iii) Use the grid to plot a graph of mean  $F$  (vertical axis) against  $d$  (in mm, on the horizontal axis) and draw a line of best fit. **Error bars should not be included.** [4]



(iv) Use your graph to determine:

I.  $W_B$ , the weight of the metal block;

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II.  $W_R$ , the weight of the ruler.

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(d) The distances,  $d$ , were measured to a resolution of  $\pm 1$  mm. Use data from the table of results to show that the percentage uncertainty in mean  $F$  is greater than the percentage uncertainty in  $d$ , when  $d = 100$  mm. [3]

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(e) Suggest **one** change to his method which would improve the quality of Sam's measurements.

[1]

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